

Notes: Beck, Levine, and Loayza (JFE, 2000)

Finance and the Sources of Growth

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1 Big picture

- This paper asks whether financial intermediary development affects long-run economic growth mainly by raising savings and capital accumulation, or instead by improving resource allocation and total factor productivity (TFP) growth.
- The core empirical message is Schumpeterian: better financial intermediaries are associated with faster GDP per capita growth primarily through faster productivity growth, not through a robust increase in private saving or physical capital accumulation.
- The central financial development variable is **Private Credit**: credit by deposit money banks and other financial institutions to the private sector divided by GDP.
- The paper uses two complementary identification approaches: cross-country instrumental variables using legal origins, and dynamic panel GMM using internal instruments.
- The paper is important because it shifts the finance-growth debate from “does finance predict growth?” to “through which source of growth does finance operate?”

2 One-sentence summary

This note summarizes Beck, Levine, and Loayza (2000), which uses cross-country IV regressions based on legal-origin instruments and dynamic panel GMM on 1960-1995 country data to show that financial intermediary development raises long-run GDP growth mainly through total factor productivity growth, while its links to physical capital accumulation and private saving are much less robust.

3 Research question

- The paper studies the empirical relation between financial intermediary development and four outcomes:
 1. real per capita GDP growth;
 2. total factor productivity growth;
 3. physical capital per capita growth;
 4. private saving rates.
- The deeper conceptual question is whether finance matters because it mobilizes more savings and capital, or because it improves allocation, monitoring, risk management, and technological/productivity growth.

4 Incremental contribution of the paper

- **Contribution relative to King and Levine (1993).** Earlier work shows that financial development predicts growth. This paper decomposes growth into productivity growth, capital accumulation, and savings to identify the channel.
- **Contribution relative to simple cross-country growth regressions.** The paper addresses simultaneity and omitted-variable concerns using two approaches: legal origin as an external instrument and dynamic panel GMM with internal instruments.

- **Contribution to growth accounting.** Instead of treating the Solow residual as unexplained, the paper directly tests whether cross-country differences in financial development help explain differences in TFP growth.
- **Contribution to financial development measurement.** The paper improves the financial development proxy by focusing on credit to the private sector from financial intermediaries, excluding central-bank and public-sector credit.
- **Economic contribution.** The findings support the idea that financial systems matter not simply because they provide more funds, but because they improve the allocation of funds toward more productive uses.

5 Economic mechanism

5.1 Schumpeterian productivity channel

- Financial intermediaries screen firms, monitor managers, gather information, manage risk, and allocate society's savings across projects.
- If these functions improve, funds move toward higher-return firms and technologies.
- This should raise productivity growth and technological progress, even if aggregate saving does not rise much.

5.2 Capital-accumulation channel

- A competing view emphasizes that finance raises growth by increasing savings, investment, and physical capital accumulation.
- In this paper, the evidence for this channel is weaker: some regressions show a positive relation with capital growth, but the result is sensitive to the financial development measure and specification.

5.3 Saving channel

- The theoretical effect of finance on saving is ambiguous.
- Better finance can raise returns, diversify risk, and loosen constraints, but income and substitution effects can move savings in different directions.
- The paper does not find robust evidence that financial intermediary development substantially raises private saving rates.

6 Data and variable construction

6.1 Sample

- Cross-country growth sample: 63 countries, data averaged over 1960-1995.
- Panel growth sample: 77 countries, seven non-overlapping five-year periods from 1960 to 1995.
- Saving sample: slightly shorter because private saving data are available mainly for 1971-1995.

6.2 Financial development measures

- **Private Credit:** credit by deposit money banks and other financial institutions to the private sector divided by GDP. This is the preferred measure.
- **Liquid Liabilities:** liquid liabilities of the financial system divided by GDP. This measures the size of the financial system.
- **Commercial-Central Bank:** commercial bank assets divided by commercial bank plus central bank assets. This captures whether commercial intermediaries, rather than the central bank, allocate credit.

6.3 Growth and sources of growth

- **Growth:** real per capita GDP growth.
- **Capgrowth:** growth in physical capital per capita.
- **Prod:** productivity growth, computed as

$$\text{Prod} = \text{Growth} - 0.3 \times \text{Capgrowth}.$$

- **Private Saving:** gross private saving divided by gross private disposable income.

7 Identification and empirical strategy

7.1 Cross-country IV design

- The cross-country regression is:

$$Y_i = \alpha + \beta \text{Finance}_i + \gamma' X_i + \varepsilon_i,$$

where Y_i is growth, capital growth, productivity growth, or private saving.

- The endogeneity problem is that financial development may respond to growth, rather than cause it.
- The instrument is **legal origin**: English, French, German, or Scandinavian legal origin.
- The exclusion logic is that legal origin is historically predetermined through colonization or occupation and affects financial development through creditor rights, contract enforcement, and disclosure rules.
- The key test is the Hansen overidentification test. Failure to reject supports the instrument set.

7.2 Dynamic panel GMM design

- The panel regression uses five-year averages and includes country-specific and time-specific effects:

$$y_{i,t} = \alpha' X_{i,t-1}^1 + \beta' X_{i,t}^2 + \mu_i + \lambda_t + \varepsilon_{i,t}.$$

- The authors use system GMM, combining equations in differences and levels.
- Lagged levels are used as instruments for differenced equations, and lagged differences are used as instruments for levels equations.

- This strategy addresses country fixed effects, weak endogeneity, and dynamic feedback.
- Specification checks include the Sargan test of instrument validity and the second-order serial-correlation test.

8 Main findings

8.1 Finance and GDP growth

- Private Credit is positively and significantly related to real per capita GDP growth in both cross-country IV and panel GMM regressions.
- The coefficient is economically meaningful. A moderate increase in Private Credit is associated with a nontrivial increase in annual growth.
- This result is robust across alternative conditioning information sets and alternative financial development measures.

8.2 Finance and productivity growth

- The strongest and most robust result is the positive relation between financial intermediary development and productivity growth.
- This supports the interpretation that finance improves the quality of capital allocation and the pace of technological progress.
- The productivity result survives both legal-origin IV and panel GMM estimation.

8.3 Finance and capital growth

- Private Credit is positively related to capital growth in the main regressions.
- However, the result is less robust when alternative financial development measures are used, especially in the pure cross-country specification.
- The authors interpret the capital-growth evidence as suggestive but not decisive.

8.4 Finance and private saving

- The private saving results are weak and inconsistent.
- Private Credit is significant in the cross-country saving regression but not in the panel regression.
- Alternative measures of financial development do not deliver a uniform positive saving result.
- The paper concludes that the finance-growth link does not primarily operate through higher private saving.

9 Table interpretations

9.1 Table 1: Summary statistics and correlations

Read: Panel A shows substantial cross-country variation in financial development and growth outcomes. Private Credit ranges from very low values to above 100 percent of GDP. Panel B shows

Table 1
Summary statistics
Private Credit is credit by deposit money banks and other financial institutions to the private sector divided by GDP. Liquid Liabilities is liquid liabilities of the financial system, calculated as currency plus demand and interest-bearing liabilities of banks and nonbank financial intermediaries, divided by GDP. Commercial-Central Bank is assets of deposit money banks divided by deposit money bank plus central bank assets. These three variables are constructed using data from the International Financial Statistics. Economic Growth is the growth rate of real per capita GDP. GDP data are from Loayza et al. (1998). Capital Growth is the growth rate of physical capital per capita and is constructed using data from PWT 5.6. Productivity Growth is Economic Growth – 0.3* Capital Growth. Private Saving is the ratio of gross private saving and gross private disposable income. Data on Private Saving are from Loayza et al. (1998). The data are averaged over the period 1960–1995, with the exception of Private Saving, for which data are averaged over the period 1971–1995. The statistics for private saving and its correlation with the three measures of financial intermediary development are from a different sample. P-values are reported under the respective coefficient.

Panel A: Descriptive Statistics							
	Private Credit	Liquid Liabilities	Commercial-Central Bank	Economic Growth	Capital Growth	Productivity Growth	Private Saving
Mean	40.86	45.21	79.26	1.95	3.13	1.01	19.21
Median	27.81	41.02	83.89	1.98	3.11	1.15	19.98
Maximum	141.30	143.43	98.89	7.16	10.51	5.14	33.92
Minimum	4.08	14.43	23.72	–	–1.84	–3.39	1.05
Std. Dev.	29.16	26.26	17.37	1.92	2.22	1.52	7.65
Observations	63	63	63	63	63	63	61

Panel B: Correlations							
Private Credit	1.00						
Liquid Liabilities	0.77	1.00					
Commercial-Central Bank	0.64	0.59	1.00				
Economic Growth	0.43	0.56	0.46	1.00			
Capital Growth	0.34	0.36	0.25	0.71	1.00		
Productivity Growth	0.39	0.55	0.47	0.95	0.46	1.00	
Private Saving	0.75	0.65	0.73	0.01	0.01	0.01	1.00

Table 1, Panel A: Summary statistics

Table 1, Panel B: Correlations

that financial development is positively correlated with GDP growth, capital growth, productivity growth, and private saving.

Economic message: The correlations are not causal evidence, but they motivate the later IV and panel tests. The strong correlation between Private Credit and productivity growth is already consistent with the paper’s main productivity-channel argument.

9.2 Tables 2 and 3: Finance, GDP growth, and productivity growth

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Table 2
Financial intermediation and economic growth

The regression equation estimated in columns 1 and 3 is $\text{Growth} = \beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Private Credit. The dependent variable is the growth rate of real per capita GDP. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Private Credit is the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in columns 2 and 4 is $\text{Growth} = \beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Private Credit. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions in columns 1 and 2 are cross-country regressions, with data averaged over 1960–1995, and using the legal origin of countries as instruments for Private Credit. The regressions in columns 3 and 4 are panel regressions, with data averaged over seven 5-year periods from 1960–1995, and using lagged values as instruments, as described in the text. The regressions in columns 3 and 4 also contain time dummies that are not reported. *P*-values calculated from White’s heteroskedasticity-consistent standard errors are reported under the respective coefficient. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values of the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation.

	Cross-country data		Panel data	
	(1)	(2)	(3)	(4)
Constant	6.571 0.006	2.643 0.527	1.272 0.250	0.082 0.875
Initial income per capita	– 1.971 0.001	– 1.967 0.001	– 1.299 0.001	– 0.496 0.001
Average years of schooling	1.936 0.008	1.548 0.078	2.671 0.001	0.950 0.001
Openness to trade		0.931 0.042		1.311 0.001
Inflation		4.270 0.096		0.181 0.475
Government size		– 1.207 0.132		– 1.445 0.001
Black market premium		– 0.139 0.914		– 1.192 0.001
Private Credit	2.215 0.003	3.215 0.012	2.397 0.001	1.443 0.001
Hansen test	0.577	0.571		
Sargan test (<i>p</i> -value)			0.183	0.506
Serial correlation test (<i>p</i> -value)			0.516	0.803
Countries	63	63	77	77
Observations			365	365

Table 2: Financial intermediation and economic growth

Read: Table 2 shows that Private Credit is positive and statistically significant in both cross-country and panel growth regressions. Table 3 shows the same pattern for productivity growth.

Economic message: These are the core results. Finance is not merely correlated with income growth; after addressing simultaneity and fixed effects, financial development remains strongly linked to growth and productivity. The productivity table is especially important because it identifies the main channel.

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Table 3
Financial intermediation and productivity growth

The regression equation estimated in columns 1 and 3 is $\text{Productivity Growth} = \beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Private Credit. The dependent variable is Productivity Growth, which is growth of real per capita GDP minus 0.3* growth of capital per capita. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Private Credit is the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in columns 2 and 4 is $\text{Productivity Growth} = \beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Private Credit. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions in columns 1 and 2 are cross-country regressions, with data averaged over 1960–1995, and using the legal origin of countries as instruments for Private Credit. The regressions in columns 3 and 4 are panel regressions, with data averaged over seven 5-year periods from 1960–1995, and using lagged values as instruments, as described in the text. The regressions in columns 3 and 4 also contain time dummies that are not reported. *P*-values calculated from White’s heteroskedasticity-consistent standard errors are reported under the respective coefficient. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation.

	Cross-country data		Panel data	
	(1)	(2)	(3)	(4)
Constant	3.527 0.065	– 1.189 0.717	2.473 0.001	– 1.611 0.033
Initial income per capita	– 1.266 0.001	– 1.171 0.001	– 1.244 0.001	– 0.353 0.001
Average years of schooling	1.375 0.028	1.241 0.060	3.043 0.001	1.174 0.001
Openness to trade		0.956 0.015		1.337 0.001
Inflation		3.223 0.096		– 0.415 0.033
Government size		– 0.647 0.286		– 0.431 0.088
Black market premium		– 0.191 0.861		– 1.003 0.001
Private Credit	1.500 0.004	1.986 0.021	1.332 0.001	0.296 0.001
Hansen test	2.036	3.472		
Sargan test (<i>p</i> -value)			0.205	0.401
Serial correlation test (<i>p</i> -value)			0.772	0.865
Countries	63	63	77	77
Observations			365	365

Table 3: Financial intermediation and productivity growth

Table 4
Financial intermediation and capital growth

The regression equation estimated in columns 1 and 3 is Capital Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Private Credit. The dependent variable is the growth rate of physical per capita capital. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Private Credit is the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in columns 2 and 4 is Capital Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Private Credit. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions in columns 1 and 2 are cross-country regressions, with data averaged over 1960–1995, and using the legal origin of countries as instruments for Private Credit. The regressions in columns 3 and 4 are panel regressions, with data averaged over seven 5-year periods from 1960–1995, and using lagged values as instruments, as described in the text. The regressions in columns 3 and 4 also contain time dummies that are not reported. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported under the respective coefficient. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation.

	Cross-country data		Panel data	
	(1)	(2)	(3)	(4)
Constant	8.448 0.004	8.349 0.093	−1.273 0.219	5.694 0.001
Initial income per capita	−2.075 0.001	−2.225 0.001	−0.933 0.001	−0.070 0.701
Average years of schooling	0.663 0.427	0.628 0.559	0.985 0.055	−0.340 0.552
Openness to trade		0.245 0.663		−0.448 0.097
Inflation		4.196 0.236		0.445 0.360
Government size		−1.619 0.082		−3.229 0.001
Black market premium		0.304 0.826		−0.748 0.001
Private Credit	2.832 0.006	4.038 0.012	3.435 0.001	3.005 0.001
Hansen test	6.747	3.039		
Sargan test (<i>p</i> -value)			0.166	0.316
Serial correlation test (<i>p</i> -value)			0.014	0.053
Countries	63	63	77	77
Observations			365	365

Table 4: Financial intermediation and capital growth

Table 5
Alternative measures of financial intermediary development and growth, using cross-country data

The regression equation estimated in Panel A is Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Finance. The dependent variable is the growth rate of real per capita GDP. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Finance is either Liquid Liabilities, the log of liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, the log of assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in Panel B is Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Finance. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions are cross-country regressions, with data averaged over 1960–1995, and using the legal origin of countries as instruments for Finance. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. There are 63 countries included in the sample.

Financial variable	Coefficient	<i>p</i> -value	Hansen test
<i>Panel A: Regressions using the simple conditioning information set</i>			
Liquid Liabilities	1.667	0.023	1.553
Commercial Central Bank	10.169	0.001	1.403
Private Credit	2.215	0.003	0.577
<i>Panel B: Regressions using the policy conditioning information set</i>			
Liquid Liabilities	2.173	0.020	2.393
Commercial Central Bank	9.641	0.021	2.350
Private Credit	3.215	0.012	0.571

do not have a significant impact on capital per capita growth in the cross-country sample. In the panel estimations, all three financial intermediary development indicators are associated with faster capital per capita growth. However, only in the regressions with Commercial-Central Bank is the null hypothesis of no serial correlation in the error term not rejected in all the specifications. Thus, while evidence suggests that financial intermediary development positively influences physical capital accumulation, the pure, cross-sectional relation between physical capital growth and financial intermediary development is highly dependent on the measure of financial intermediary development used.

Table 5: Alternative financial measures and growth, cross-country data

9.3 Tables 4 and 5: Capital growth and alternative finance-growth measures

Read: Table 4 suggests that Private Credit is positively related to capital growth, but the specification tests and later robustness checks make this channel less clean. Table 5 shows that alternative financial development measures also predict GDP growth in cross-country IV regressions.

Economic message: The overall growth relation is robust, but the physical-capital channel is weaker than the productivity channel. This distinction is central to the paper's contribution.

9.4 Tables 6 and 7: Alternative financial development measures

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Table 6

Alternative measures of financial intermediary development and growth, using panel data

The regression equation estimated in Panel A is $\text{Growth} = \beta_0 + \beta_1 \text{Initial income per capita} + \beta_2 \text{Average years of schooling} + \beta_3 \text{Finance}$. The dependent variable is the growth rate of real per capita GDP. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Finance is either Liquid Liabilities, the log of liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, the log of assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated Panel B is $\text{Growth} = \beta_0 + \beta_1 \text{Initial income per capita} + \beta_2 \text{Average years of schooling} + \beta_3 \text{Openness to trade} + \beta_4 \text{Inflation} + \beta_5 \text{Government size} + \beta_6 \text{Black market premium} + \beta_7 \text{Finance}$. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions are panel regressions, with data averaged over seven 5-year periods from 1960–1995, and using lagged values as instruments, as described in the text. The regressions also contain time dummies. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation. There are 77 countries and 365 observations included in the sample.

Financial variable	Coefficient	<i>p</i> -value	Sargan set (<i>p</i> -value)	2nd order serial corr. test (<i>p</i> -value)
<i>Panel A: Regressions using the simple conditioning information set</i>				
Liquid Liabilities	2.093	0.001	0.227	0.522
Commercial Central Bank	4.763	0.001	0.246	0.712
Private Credit	2.397	0.001	0.183	0.516
<i>Panel B: Regressions using the policy conditioning information set</i>				
Liquid Liabilities	2.321	0.001	0.607	0.722
Commercial Central Bank	3.361	0.001	0.390	0.958
Private Credit	1.443	0.001	0.506	0.803

5. Finance and private saving

This section explores the impact of the exogenous component of financial development on private savings rates. As in the previous section, we will use both cross-country and panel samples, but a different set of conditioning information.

Table 6: Alternative financial measures and growth, panel data

Read: Table 6 confirms that Liquid Liabilities, Commercial-Central Bank, and Private Credit are all positively related to growth in panel regressions. Table 7 shows similar robustness for productivity growth in cross-country regressions.

Economic message: These results reduce the concern that the findings are driven only by the specific Private Credit measure. The finance-growth and finance-productivity links are broadly robust.

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Table 7

Alternative measures of financial intermediary development and productivity growth, using cross-country data

The regression equation estimated in Panel A is $\text{Productivity Growth} = \beta_0 + \beta_1 \text{Initial income per capita} + \beta_2 \text{Average years of schooling} + \beta_3 \text{Finance}$. The dependent variable is Productivity Growth, which is growth of real per capita GDP minus 0.3* growth of capital per capita. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Finance is either Liquid Liabilities, the log of liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, the log of assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in Panel B is $\text{Productivity Growth} = \beta_0 + \beta_1 \text{Initial income per capita} + \beta_2 \text{Average years of schooling} + \beta_3 \text{Openness to trade} + \beta_4 \text{Inflation} + \beta_5 \text{Government size} + \beta_6 \text{Black market premium} + \beta_7 \text{Finance}$. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions are cross-country regressions, with data averaged over 1960–1995, and using the legal origin of countries as instruments for Finance. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. There are 63 countries included in the sample.

Financial variable	Coefficient	<i>p</i> -value	Hansen test
<i>Panel A: Regressions using the simple conditioning information set</i>			
Liquid Liabilities	1.787	0.002	0.253
Commercial Central Bank	5.853	0.001	0.092
Private Credit	1.500	0.004	2.036
<i>Panel B: Regressions using the policy conditioning information set</i>			
Liquid Liabilities	2.168	0.006	3.315
Commercial Central Bank	8.134	0.006	1.284
Private Credit	1.986	0.021	3.472

5.1. The conditioning information set

The set of conditioning information is selectively determined by various theories of consumption, including the classical permanent-income and life-cycle hypotheses and the more recent theories accounting for consumption habits, subsistence consumption, precautionary saving motives, and borrowing constraints (see Loayza et al., 2000). The variables included in the set of conditioning information for the saving regression are listed below.

Table 7: Alternative financial measures and productivity growth, cross-country data

Table 8

Alternative measures of financial intermediary development and productivity growth, using panel data

The regression equation estimated in Panel A is Productivity Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Finance. The dependent variable is Productivity Growth, which is growth of real per capita GDP minus 0.3* growth of capital per capita. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Finance is either Liquid Liabilities, the log of liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, the log of assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in Panel B is Productivity Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Finance. Openness to trade is the log of the sum of real exports and imports of goods and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions are panel regressions, with data averaged over seven 5-year periods from 1960–1995, and using lagged values as instruments, as described in the text. The regressions also contain time dummies. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation. There are 77 countries and 365 observations included in the sample.

Financial variable	Coefficient	<i>p</i> -value	Sargan set (<i>p</i> -value)	2nd order serial corr. test (<i>p</i> -value)
<i>Panel A: Regressions using the simple conditioning information set</i>				
Liquid Liabilities	0.663	0.001	0.124	0.841
Commercial Central Bank	2.388	0.001	0.242	0.965
Private Credit	1.332	0.001	0.205	0.772
<i>Panel B: Regressions using the policy conditioning information set</i>				
Liquid Liabilities	0.856	0.001	0.552	0.836
Commercial Central Bank	1.669	0.001	0.486	0.758
Private Credit	0.296	0.001	0.401	0.865

The level and the growth rate of private income have ambiguous effects on saving regressions, depending on whether the change in these variables is permanent or temporary. These ambiguous effects also occur depending on whether the change takes place within a generation or across generations. The same argument holds for the terms of trade, which can be considered an exogenous determinant of income. The level of income may have an additional,

Table 8: Alternative financial measures and productivity growth, panel data

Table 9

Alternative measures of financial intermediary development and capital growth, using cross-country data

The regression equation estimated in Panel A is Capital Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Finance. The dependent variable is the growth rate of physical per capita capital. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Finance is either Liquid Liabilities, the log of liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, the log of assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in Panel B is Capital Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Finance. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions are cross-country regressions, with data averaged over 1960–1995, and using the legal origin of countries as instruments for Finance. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. There are 63 countries included in the sample.

Financial variable	Coefficient	<i>p</i> -value	Hansen test
<i>Panel A: Regressions using the simple conditioning information set</i>			
Liquid Liabilities	− 0.345	0.767	4.693
Commercial Central Bank	− 1.046	0.832	4.578
Private Credit	2.832	0.006	6.747
<i>Panel B: Regressions using the policy conditioning information set</i>			
Liquid Liabilities	0.511	0.562	4.605
Commercial Central Bank	1.018	0.755	4.722
Private Credit	4.038	0.012	3.039

positive impact on the private savings rate if a large share of the country's population is near subsistence consumption levels. Government saving, expressed relative to GPDI in our saving regressions, is another important variable, serving to account for Ricardian equivalence effects. The expected sign for government saving is negative, reflecting at least a partial private saving offset of changes in public saving. We include a measure of the real interest rate, which has well known negative substitution and positive income effects on consumption, resulting in an ambiguous sign in saving regressions. We include the inflation rate as a proxy for uncertainty, expecting a positive association

Table 9: Alternative financial measures and capital growth, cross-country data

9.5 Tables 8 and 9: Productivity and capital-growth robustness

Read: Table 8 shows that alternative financial development measures predict productivity growth in panel regressions. Table 9 shows that alternative financial measures do not consistently predict capital growth in cross-country regressions.

Economic message: The contrast between Tables 8 and 9 is one of the cleanest pieces of evidence in the paper. Productivity is the robust channel; capital accumulation is not.

9.6 Tables 10 and 11: Capital growth in panel data and private saving

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Table 10

Alternative measures of financial intermediary development and capital growth, using panel data

The regression equation estimated in Panel A is Capital Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Finance. The dependent variable is the growth rate of physical per capita capital. Initial income per capita is the log of real per capita GDP in the first year of the respective time period. Average years of schooling is log of one plus the average years of schooling in the total population over 25. Finance is either Liquid Liabilities, the log of liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, the log of assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, the log of credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regression equation estimated in Panel B is Capital Growth = $\beta_0 + \beta_1$ Initial income per capita + β_2 Average years of schooling + β_3 Openness to trade + β_4 Inflation + β_5 Government size + β_6 Black market premium + β_7 Finance. Openness to trade is the log of the sum of real exports and imports of goods and nonfinancial services as share of real GDP. Inflation is the log of one plus the inflation rate, calculated using the average annual CPI data from the International Financial Statistics. Government size is the log of real general government consumption as share of real GDP. Black market premium is the log of one plus the black market premium. The regressions are panel regressions, with data averaged over seven 5-year periods from 1960–1995, and using lagged values as instruments, as described in the text. The regressions also contain time dummies. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation. There are 77 countries and 365 observations included in the sample.

Financial variable	Coefficient	<i>p</i> -value	Sargan set (<i>p</i> -value)	2nd order serial corr. test (<i>p</i> -value)
<i>Panel A: Regressions using the simple conditioning information set</i>				
Liquid Liabilities	3.667	0.001	0.192	0.013
Commercial Central Bank	8.848	0.001	0.258	0.172
Private Credit	3.435	0.001	0.166	0.014
<i>Panel B: Regressions using the policy conditioning information set</i>				
Liquid Liabilities	5.162	0.001	0.494	0.076
Commercial Central Bank	6.493	0.001	0.338	0.169
Private Credit	3.005	0.001	0.316	0.053

between saving and the inflation rate, consistent with a precautionary saving motive. This result would only be the partial effect of inflation on saving. The net effect of inflation on saving would also consider the negative effect of inflation on, among other variables, income growth.

We include several demographic variables. The first are the old-age and young-age population dependency ratios, defined, respectively, as the ratios of population under 15 years of age and over 65 year of age to total population. Including the dependency ratios helps account for life-cycle effects. The standard

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Table 11

Financial intermediation and private saving

The regression equation estimated is Private Saving = $\beta_0 + \beta_1$ Real per capita GDP + β_2 Growth rate of real per capita GDP + β_3 Real interest rate + β_4 Terms of trade + β_5 Old dependency ratio + β_6 Young dependency ratio + β_7 Urbanization ratio + β_8 Government Saving + β_9 Inflation + β_{10} Private Credit. The dependent variable is Private Saving which is the ratio of gross private saving and gross private disposable income. Real per capita GDP is the log of real per capita gross private disposable income. Real interest rate is the log of one plus the real interest rate. Terms of trade is the log of the ratio of export and import prices. Old dependency ratio is the share of population over 65 in total population. Young dependency ratio is the share of population under 15 in total population. Urbanization ratio is the share of population that lives in urban areas. Government Saving is the ratio of gross public saving and gross private disposable income. Inflation is the log of one plus the inflation rate. Private Credit is credit by deposit money banks and other financial institutions to the private sector as share of GDP. The regression in column 1 is a cross-country regression, with data averaged over 1971–1995, and using the legal origin of countries as instruments for Private Credit. The regression in column 2 is a panel regression, with data averaged over five 5-year periods from 1971–1995, and using lagged values as instruments, as described in the text. The regression in column 2 also contains time dummies that are not reported. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported under the respective coefficient. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation.

	(1)	(2)
Constant	−0.102 0.387	0.474 0.001
Real per capita GDP	0.041 0.005	0.000 0.992
Growth rate of real per capita GDP	1.378 0.001	0.531 0.001
Real interest rate	0.172 0.282	−0.101 0.130
Terms of trade	−0.024 0.534	−0.029 0.094
Old dependency ratio	−0.313 0.170	−0.940 0.001
Young dependency ratio	0.012 0.884	−0.300 0.001
Urbanization ratio	−0.073 0.054	0.107 0.010
Government Saving	−0.129 0.527	−0.273 0.001
Inflation	0.039 0.733	−0.327 0.001
Private Credit	0.085 0.027	0.021 0.224
Hansen test	0.708	
Sargan test (<i>p</i> -value)		0.311
Serial correlation test (<i>p</i> -value)		0.335
Countries	61	72
Observations		247

Table 10: Alternative financial measures and capital growth, panel data

Table 11: Financial intermediation and private saving

Read: Table 10 gives stronger panel evidence that finance is related to capital growth, but some serial-correlation concerns remain. Table 11 shows that Private Credit is significant for saving in cross-section but not in panel data.

Economic message: The results are consistent with a possible short-run or panel relation between finance and investment, but they do not establish a robust long-run saving channel.

Table 12

Alternative measures of financial intermediary development and private saving

The regression equation estimated is Private Saving = $\beta_0 + \beta_1$ Real per capita GDP + β_2 Growth rate of real per capita GDP + β_3 Real interest rate + β_4 Terms of trade + β_5 Old dependency ratio + β_6 Young dependency ratio + β_7 Urbanization ratio + β_8 Government Saving + β_9 Inflation + β_{10} Finance. The dependent variable is Private Saving which is the ratio of gross private saving and gross private disposable income. Real per capita GDP is the log of real per capita gross private disposable income. Real interest rate is the log of one plus the real interest rate. Terms of trade is the log of the ratio of export and import prices. Old dependency ratio is the share of population over 65 in total population. Young dependency ratio is the share of population under 15 in total population. Urbanization ratio is the share of urban population that lives in urban areas. Government Saving is the ratio of gross public saving and gross private disposable income. Inflation is the log of one plus the inflation rate. Finance is either Liquid Liabilities, liquid liabilities of the financial system divided by GDP, Commercial-Central Bank, assets of deposit money banks divided by deposit money bank plus central bank assets, or Private Credit, credit by deposit money banks and other financial institutions to the private sector divided by GDP. The regressions in Panel A are cross-country regressions, with data averaged over 1971–1995, and using the legal origin of countries as instruments for Finance. The regressions in Panel B are panel regressions, with data averaged over five 5-year periods from 1971–1995, and using lagged values as instruments, as described in the text. The regressions in panel B also contain time dummies. *P*-values calculated from White's heteroskedasticity-consistent standard errors are reported. The null hypothesis of the Hansen test is that the instruments used are not correlated with the residuals. The critical values for the Hansen test (2 d.f.) are: 10% = 4.61; 5% = 5.99. The null hypothesis of the Sargan test is that the instruments used are not correlated with the residuals. The null hypothesis of the serial correlation test is that the errors in the first-difference regression exhibit no second-order serial correlation.

Panel A: Cross-country data

Financial variable	Coefficient	<i>p</i> -value	Hansen test
Liquid Liabilities	0.075	0.102	3.106
Commercial Central Bank	0.896	0.338	1.370
Private Credit	0.085	0.027	0.708

Panel B: Panel data

Financial variable	Coefficient	<i>p</i> -value	Sargan test (<i>p</i> -value)	2nd order serial corr. test (<i>p</i> -value)
Liquid Liabilities	− 0.012	0.588	0.631	0.286
Commercial Central Bank	0.154	0.001	0.363	0.340
Private Credit	0.021	0.224	0.311	0.335

We find an economically large and statistically significant relation between financial intermediary development and both real per capita GDP growth and total factor productivity growth. Specification tests indicate that the robust, positive relation between financial development and both growth and productivity growth are not due to simultaneity bias or country-specific effects. This

Table 12: Alternative measures of financial intermediary development and private saving

9.7 Table 12: Alternative financial development measures and private saving

Read: The saving results differ by financial measure and estimation method. Private Credit is significant only in cross-section; Commercial-Central Bank is significant in panel data, while other measures are not consistently significant.

Economic message: There is no stable evidence that financial intermediary development raises private saving. This weakens capital-fundamentalist interpretations of the finance-growth nexus.

10 How to read the empirical design

10.1 Why legal origin is used

- Legal origin affects creditor rights, contract enforcement, and accounting standards.
- These institutions affect how effectively financial intermediaries can lend to private firms.
- Because legal origin was largely inherited through colonial and historical processes, the authors treat it as plausibly exogenous to post-1960 growth outcomes.

10.2 Why panel GMM is used

- Cross-country IV identifies long-run cross-sectional variation, but may omit country-specific effects.
- Panel GMM uses within-country changes and controls for unobserved country effects.
- The method also treats regressors as potentially endogenous and instruments them with their own lags.

10.3 Why the two approaches are complementary

- Cross-country IV is closer to long-run development comparisons.
- Panel GMM uses more variation but may mix long-run growth with medium-run cyclical dynamics.
- When both methods produce the same result, the evidence is much stronger.

11 Interpretation and intuition

- If finance mainly worked through savings, we would expect robust positive coefficients in saving regressions. The paper does not find this.
- If finance mainly worked through capital accumulation, we would expect all financial development measures to robustly predict capital growth. The evidence is mixed.
- If finance works through allocation and innovation, we should see robust productivity effects. This is exactly what the paper finds.
- The paper therefore interprets financial intermediaries as institutions that improve selection, monitoring, information production, and resource allocation.

12 Relation to the literature

- **Schumpeter (1934):** Finance affects economic development by selecting entrepreneurs and allocating credit.
- **King and Levine (1993):** Finance predicts growth; this paper studies the channels behind that relation.
- **Rajan and Zingales (1998):** Finance affects industries that depend more on external finance; this paper uses country-level data and growth-accounting outcomes.
- **Levine, Loayza, and Beck (2000):** Closely related causal finance-growth evidence; this paper focuses specifically on sources of growth.
- **Growth accounting literature:** The paper contributes to explanations of TFP differences across countries.

13 Limitations

- Legal origin may affect growth through institutional channels other than financial intermediary development.
- Dynamic panel GMM relies on internal instruments and weak-exogeneity assumptions that may be imperfect in small samples.
- Private Credit is a useful but still indirect measure of financial services such as screening, monitoring, and risk management.
- TFP is measured as a residual, so measurement error in capital or output can affect productivity results.
- The paper focuses on intermediaries and does not fully incorporate securities market development due to data limitations.

14 Takeaway

- The main takeaway is that finance matters for growth mainly because it improves productivity, not because it robustly raises savings or capital accumulation.
- This makes financial development a resource-allocation mechanism: better financial systems help economies select better projects, monitor firms more effectively, and move capital toward more productive uses.
- For empirical research, the paper is a useful benchmark for combining external instruments, dynamic panel GMM, and channel decomposition.